

Neural ODE Research — Literature & Experimental Validation

Tags

Research Validation Phase

Common Theme

Neural Ordinary Differential Equations for Stiff and Non-Stiff Dynamical Systems

1. Purpose of This Phase

The objective of this phase is to investigate whether there are **well-defined and publishable research questions** within the use of Neural ODEs for scientific computation.

You will work on the same broad domain of **stiff and non-stiff ODEs**, but you will investigate **different research questions**.

At this stage, you are **not expected to develop a new algorithm**.

Instead, you must:

1. Understand the existing literature.
2. Identify what has already been established.
3. Identify a specific limitation or unanswered question.
4. Formulate a testable hypothesis.
5. Design controlled experiments.
6. Perform preliminary experiments.
7. Determine whether the hypothesis is sufficiently supported to justify further research.

Important: The proposed research direction is provisional. The final research problem will be fixed only after the literature and preliminary experiments have been evaluated.

2. Common Research Context

Neural ODEs provide a continuous formulation of neural dynamics:

$$\frac{dz}{dt}(t) = f_{\theta}(z(t), t)$$

However, the model is ultimately trained and executed using **numerical methods**.

Therefore, the practical system involves:

Continuous Dynamics → Neural Approximation → Numerical Computation → Computer Hardware

This raises several questions about how the numerical and computational realization affects Neural ODEs.

The two research tracks below investigate **different parts of this pipeline**.

3. Research Track 1 — Numerical Realization

Provisional Topic

Numerical Realization and Learned Dynamics in Neural ODEs

Research Question

How does the numerical realization used during Neural ODE training affect the dynamics actually learned, and does this sensitivity change across stiff and non-stiff regimes?

Initial Hypothesis

The sensitivity of learned Neural ODE dynamics to numerical realization varies systematically with dynamical difficulty/stiffness and cannot be fully explained by final trajectory error alone.

This hypothesis is **not established**. Your job is to determine whether the literature and experiments support it.

3.1 Literature Investigation

Study existing work on:

- Neural ODEs
- numerical integration of Neural ODEs
- solver-dependent training
- discretization effects
- numerical stability
- stiff Neural ODEs
- implicit Neural ODE methods
- inverse modified differential equations
- numerical error and learning error

Minimum requirement

Identify:

- **15 relevant papers**
- **5 closest papers studied in depth**

For each paper record:

Paper	Problem	Method	ODE/System	Metrics	Main Finding	Limitation	Relation Question
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3.2 Literature Questions

You must determine:

1. What is already known about solver-dependent Neural ODE learning?
2. How has numerical discretization been shown to affect learned dynamics?
3. What is already known about stiff Neural ODE training?
4. Has anyone already studied numerical sensitivity specifically as a function of stiffness?
5. What metrics have previous studies used?
6. What important aspects have previous studies not investigated?
7. Are there papers already addressing our proposed hypothesis directly?

Do **not** claim that something is novel merely because you did not find it in the first search.

3.3 Initial Experimental Study

Begin with a small benchmark set containing:

- at least **2 non-stiff systems**
- at least **2 stiff systems**

Where possible, investigate a progression such as:

Low stiffness → Moderate stiffness → High stiffness

Use established Neural ODE architectures and numerical methods initially.

3.4 Controlled Numerical Experiments

Keep the model and data fixed while changing one numerical factor at a time:

- solver
- step size
- tolerance
- solver order
- training solver
- evaluation solver

Initially, **do not develop a new solver**.

3.5 Measurements

Investigate:

- trajectory error
- vector-field difference where meaningful
- stability
- long-term prediction
- NFE
- conservation/invariant violation where applicable
- sensitivity to changing the numerical solver during evaluation

A potential quantity to investigate is:

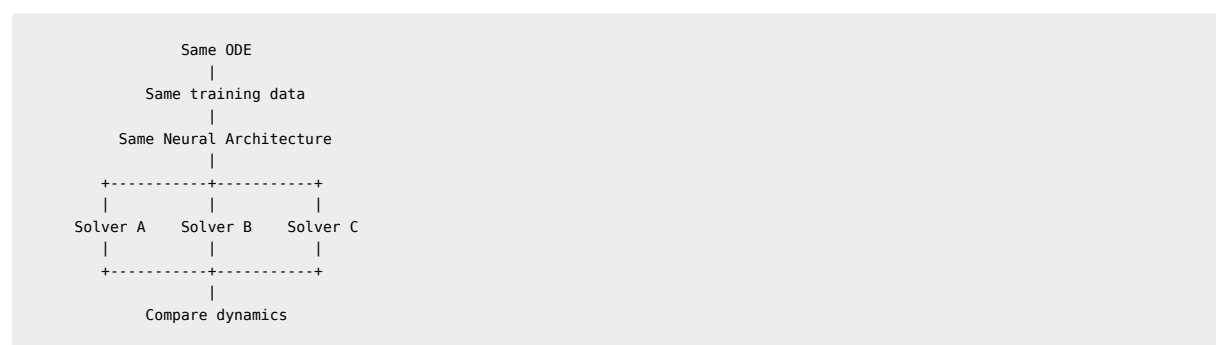
$$S_{\text{num}} = D(\Phi_{\theta}(A), \Phi_{\theta}(B))$$

where D represents an appropriate measure of dynamical difference.

The final definition must be justified from the literature.

3.6 Critical Experiment

For the same ODE:



Repeat this across different stiffness regimes.

3.7 Final Question for Track 1

Determine whether:

S_{num}

has a meaningful relationship with:
stiffness / dynamical difficulty
and whether numerical sensitivity reveals information that cannot be captured by ordinary trajectory error alone.

4. Research Track 2 — Computational Realization

Provisional Topic

Computational Performance Characterization of Neural ODEs

Research Question

How do the numerical characteristics of a Neural ODE translate into actual computational cost on modern hardware, and is NFE sufficient to characterize that cost?

Initial Hypothesis

NFE alone is insufficient to characterize the computational cost of Neural ODEs across different numerical regimes, model sizes, and hardware configurations.

Again, this is a hypothesis to be tested.

4.1 Literature Investigation

Study:

- Neural ODE computational efficiency
- Neural ODE performance
- GPU-based Neural ODEs
- scientific computing implementations
- solver computational cost
- NFE as a computational metric
- GPU/CPU execution
- memory behavior
- energy efficiency
- efficient Neural ODE implementations

Identify:

- **15 relevant papers**
- **5 closest papers studied in depth**

Use the same literature matrix:

Paper	Problem	Method	Hardware	Metrics	Main Finding	Limitation	Relation Question
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4.2 Literature Questions

Determine:

1. How is Neural ODE computational cost currently measured?
2. How commonly is NFE used as the cost metric?
3. Is NFE a reliable predictor of runtime?

4. What factors besides NFE affect execution cost?
 5. How do stiff and non-stiff systems differ computationally?
 6. How does state dimension affect cost?
 7. How does neural-network size affect cost?
 8. What has already been studied on GPUs?
 9. Has anyone already developed a performance model similar to our proposed model?
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4.3 Initial Computational Model

Begin with:

$TNODE = NFE \times Tf + Tsolver + Tmemory + Tlaunch + Tsync$

This is only a starting hypothesis.

The purpose of the experiments is to determine which factors actually dominate.

4.4 Experimental Variables

Use the same core ODE benchmark set as Track 1.

Investigate:

Numerical characteristics

- stiffness
- NFE
- solver
- step count
- state dimension
- neural-network size
- fixed/adaptive stepping

Hardware

Initially:

- CPU
- GPU

Where measurement infrastructure permits:

- memory usage
 - utilization
 - power
 - energy
 - kernel behavior
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4.5 Critical Experiment

Compare workloads such as:

```
Workload A
High NFE
Small state
Small network
```

vs.

Workload B
Low NFE
Large state
Large network

Measure:

NFE

against:

Runtime

and, where possible:

Energy

Determine whether NFE alone explains the computational cost.

4.6 Hardware Scaling

After establishing the basic relationship, investigate:

CPU ↔ GPU

and, if resources permit:

- batch size
- numerical precision
- model size
- state dimension

Do not make mixed precision or GPU implementation itself the contribution.

4.7 Final Question for Track 2

Determine:

Can Neural ODE computational cost be adequately predicted from NFE, or do numerical and hardware characteristics introduce additional significant factors?

If additional factors exist, identify and quantify them.

5. Experimental Rules for Both Interns

Rule 1 — One variable at a time

Do not simultaneously change solver, architecture, precision, batch size, and hardware unless studying their interaction deliberately.

Rule 2 — Reproducibility

Every experiment must record:

Experiment ID:
Date:
ODE:
ODE parameters:
Stiffness:
Architecture:
Solver:
Tolerance:
Step size:
Training configuration:
Precision:
Hardware:

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Random seed:
NFE:
Runtime:
Memory:
Error:
Observation:
```

Rule 3 — Record negative results

A failed hypothesis is a valid research result.

Rule 4 — No premature algorithm development

Do not spend time inventing a new algorithm before establishing that the research phenomenon exists.

Rule 5 — Every graph must answer a question

Do not generate plots merely because data is available.

6. Weekly Report

Submit a **one-page report every week** containing:

1. Literature studied

Important papers read this week.

2. Research question

What question were you investigating?

3. Experiment

What exactly did you change?

4. Results

Graphs/tables.

5. Observation

What does the result tell us?

6. Hypothesis status

- Supported
- Partially supported
- Not supported
- Inconclusive

7. Next experiment

Maximum 2–3 clearly defined experiments.

7. Research Validation Report

At the end of the validation phase, submit a **5–8 page report**.

It must contain:

1. **Problem Definition**
2. **Literature Review**
3. **Closest 5 Papers**
4. **Existing Limitations**
5. **Identified Research Gap**

6. **Research Question**
7. **Hypothesis**
8. **Experimental Design**
9. **Preliminary Results**
10. **Hypothesis Evaluation**
11. **Proposed Research Contribution**
12. **Plan for Full Experimental Study**
13. **References**